U – An ontological bayesian intelligent agenetic multidisciplinary polyglot robotics system

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Chapter 1 — Dexterity via Fourier Series

A formal basis for the self-inscribing robot hand · version 2 · 22 October 2026

1.0 Origin and scope

The idea comes from a voice note recorded on 22 October 2026 while watching a Zach D Films short on Fourier drawings: a hand draws a big circle, then smaller and smaller ones, each centred on the rim of the last; when they all spin at different sizes and speeds, the pen draws the hand that drew it. The question taken from that note is whether this can be made into a measurable test of dexterity for a humanoid robot hand: can a hand, working in R^3, draw its own outline — and, through that, a copy of itself?

Version 1 of this chapter (same date) gave a proof skeleton and measurements. Version 2 keeps the idea and the parts of v1 that hold, corrects the parts that did not, and re-measures on a real five-finger hand outline. Section 1.10 lists every change.

The chapter makes five claims, each with its own status:

- **C1** Traceability. Any continuous closed curve in R^3 is the uniform limit of finite Fourier sums. Theorem 1 — proved.
- **C2** Mechanism. In 3D each harmonic is an ellipse, which is two counter-rotating circles in a plane; the chain of harmonics lives inside overlapping bounding spheres. Lemma 1 — proved.
- **C3** Realisability. A manipulator can trace a curve to tolerance ε if it can reach it, stays away from singularities along it, and has enough task-space bandwidth. Theorem 2 — proved under stated hypotheses.
- **C4** Self-inscription. A hand that satisfies C3 for its own (scaled) outline draws itself: its outline is an ε-fixed point of its draw map. Definition + Corollary 2 — proved.
- **C5** Self-replication. A self-inscribing hand can print its own shell layer by layer. Conjecture 1 — open.

1.1 Theorem 1 — traceability of closed curves in R^3

Theorem 1. *Let $\gamma : [0, 2\pi] \to R^3$ be continuous with $\gamma(0) = \gamma(2\pi)$. For every $\varepsilon > 0$ there exist N and vectors $a_0, a_k, b_k \in R^3$ $(k = 1 \dots N)$ such that the trigonometric curve*

$$\gamma_N(t) = a_0 + \Sigma_{k=1}^{N} [\, a_k \cos(kt) + b_k \sin(kt) \,]$$

satisfies $\sup_t \|\gamma(t) - \gamma_N(t)\| < \varepsilon$ (Euclidean norm).

Proof. Each coordinate γ_j $(j = 1, 2, 3)$ is a real, continuous, 2π-periodic function. By Fejér's theorem its Cesàro means $\sigma_N \gamma_j$ — trigonometric polynomials of degree $\le N$ — converge to γ_j uniformly. Choose N so that $\sup|\gamma_j - \sigma_N \gamma_j| < \varepsilon/\sqrt{3}$ for all three coordinates; then the Euclidean error is below ε. Writing $\hat{c}_k$ for the complex Fourier

coefficients of γ, the Fejér coefficients are a_k = 2(1 − k/(N+1)) Re ĉ_k and b_k = −2(1 − k/(N+1)) Im ĉ_k. ■

Why Fejér rather than plain partial sums: for a merely continuous curve the partial sums S_N need not converge uniformly (du Bois-Reymond's counterexample). For the curves that matter here they do. If γ is absolutely continuous with $\gamma' \in L^2$ — true of any piecewise-C^1 outline — then by Cauchy–Schwarz $\Sigma|\hat{c}_k| \le |\hat{c}_0| + (\Sigma\, k^2|\hat{c}_k|^2)^{\{1/2\}} (\Sigma_{k\neq 0} k^{-2})^{\{1/2\}} < \infty$, so $S_N \to \gamma$ uniformly. All measurements below use S_N.

Corollary 1 (rate). *If γ is C^1 and γ'' has bounded variation V (a piecewise-C^2 outline: straight runs and arcs whose curvature jumps), then $|\hat{c}_k| \le V / (2\pi|k|^3)$ and*

$$\sup_t \|\gamma(t) - S_N\,\gamma(t)\| \le \Sigma_{|k|>N} |\hat{c}_k| \le V / (2\pi\, N^2)$$

If instead γ has true corners (γ' of bounded variation, γ only C^0), the same argument gives O(1/N).

Proof. For a periodic function f of bounded variation, integration by parts in the Stieltjes sense gives $|\hat{f}_k| \le V(f) / (2\pi|k|)$. Apply this to f = γ'', whose coefficients are $(ik)^2\hat{c}_k$, then sum the tail using $\Sigma_{k>N} k^{-3} \le 1/(2N^2)$, counted once for each sign of k. ■

Measured: on the filleted hand outline the fitted log–log slope of error against N (60 ≤ N ≤ 400) is -2.35, against the predicted −2. With the convex wrist corners left sharp, the measured slope was −1.06, against the predicted −1. The corollary therefore says something useful: rounding the outline's corners (which a real hand already has) is what makes it cheap to draw.

1.2 Lemma 1 — one harmonic in 3D is an ellipse, and two circles

Lemma 1. *Let $\alpha, \beta \in R^3$ and $E(t) = \alpha \cos t + \beta \sin t$. Let $\sigma_1 \ge \sigma_2 \ge 0$ be the singular values of the 3×2 matrix $A = [\alpha\ \beta]$. Then:*

(i) E traces an ellipse centred at the attachment point, lying in the plane span{α, β}, with semi-axes σ_1 and σ_2, where $\sigma_1^2 + \sigma_2^2 = |\alpha|^2 + |\beta|^2$ and $\sigma_1\sigma_2 = |\alpha \times \beta|$. It degenerates to a segment when α ∥ β, and to a point when α = β = 0.

(ii) $\sigma_2 \le |E(t)| \le \sigma_1$ for all t. The ellipse lies on a sphere only when $\sigma_1 = \sigma_2$, i.e. α ⊥ β and |α| = |β| (a circle). In general it lies in the spherical shell between radii σ_2 and σ_1.

(iii) E is the sum of two circles in the same plane, rotating in opposite directions, with radii $(\sigma_1 + \sigma_2)/2$ and $(\sigma_1 - \sigma_2)/2$.

Proof. Take the reduced singular value decomposition $A = U\Sigma V^T$ with orthonormal columns u_1, u_2. V^T maps the unit circle (cos t, sin t) to itself with $t \mapsto \pm t + \phi$, so $E(t) = \sigma_1 \cos\theta\, u_1 + \sigma_2 \sin\theta\, u_2$ with $\theta = \pm t + \phi$. This gives the ellipse in (i), and $|E|^2 = \sigma_1^2 \cos^2\theta + \sigma_2^2 \sin^2\theta$ gives (ii). The invariants follow from $\mathrm{tr}(A^TA) = |\alpha|^2 + |\beta|^2$ and $\det(A^TA) = |\alpha|^2|\beta|^2 - (\alpha\cdot\beta)^2 = |\alpha \times \beta|^2$. For (iii), $\sigma_1 \cos\theta\, u_1 + \sigma_2 \sin\theta\, u_2 = \tfrac{1}{2}(\sigma_1+\sigma_2)(\cos\theta\, u_1 + \sin\theta\, u_2) + \tfrac{1}{2}(\sigma_1-\sigma_2)(\cos\theta\, u_1 - \sin\theta\, u_2)$. ■

The overlapping spheres, made exact. The voice note's picture — spheres in 3D, each smaller one overlapping its parent, with roll, pitch and yaw by the right-hand rule — survives in this form. Harmonic k's ellipse fits inside a ball of radius $\sigma_1^{(k)}$ centred on the tip of harmonic k − 1. So the pen always lies inside a chain of balls, each centred on a point that moves inside the previous one. They overlap rather than touch, as in the video frame where every circle's centre sits on its parent's rim. Each harmonic's plane has a normal n = (α × β)/|α × β| (two orientation angles: roll and pitch). Inside that plane, two counter-rotating circles spin at rate k (yaw), and the sign of k under the right-hand rule sets the direction. Version 1's semi-axis formula (|α| ± |β|)/2 and its claim that the ellipse lies on a sphere were both wrong; Lemma 1 replaces them.

A Fourier drawing machine has one degree of freedom. All the circles are driven from a single input t through fixed ratios k, and the six numbers in (α_k, β_k) are design constants, not degrees of freedom. Version 1's table — "27 DOF buys N = 8 gimballed or N = 26 spinning harmonics" — is therefore withdrawn. Joint count does not buy harmonics. It buys reconfigurability: the ability to draw any curve without rebuilding the gears. What limits how many harmonics a reconfigurable hand can draw is how fast it can follow a changing target, which is Theorem 2.

1.3 The 27-DOF accounting

The commonly cited 27-DOF kinematic model of the human hand breaks down as follows. (The v1 table differed: it gave the thumb IP joint 2 DOF, and it added palm-arch DOF and a forearm "lock-out", which is a constraint, not a DOF.)

Segment	DOF	Joints
Four fingers	4 × 4 = 16	MCP 2 (flexion, ab/adduction); PIP 1; DIP 1
Thumb	5	CMC 2; MCP 2; IP 1
Hand pose in space	6	3 rotations + 3 translations of the palm frame (supplied by wrist and arm)
Total	27	21 joints of the hand itself + 6 that place it

This split matters for drawing. The 21 joints of the hand itself move a pen only within a region about the size of a finger (≈ 8 cm in the model below), while the outline is 189 mm long. Drawing the hand at full size therefore needs the 6 pose DOF, i.e. an arm. With the wrist fixed, the hand can only draw a scaled-down copy of itself — the "smaller one" drawn at the centre in the voice note (see Corollary 2).

1.4 Theorem 2 — when a manipulator can trace a curve

Let the manipulator have joint vector $q \in Q \subset R^n$ ($n \geq 3$), pen-tip forward kinematics $f : Q \to R^3$, and Jacobian $J(q) = \partial f/\partial q$, a 3 × n matrix. Split the tolerance as $\varepsilon = \varepsilon_F + \varepsilon_C$, with Fourier error ε_F and control error ε_C. Let $N^*(\varepsilon_F)$ be the smallest N such that $\sup\|\gamma - S_N \gamma\| < \varepsilon_F$ (finite by Theorem 1), and write γ_N for $S_N \gamma$ with $N = N^*(\varepsilon_F)$.

(H1) Reach. $\gamma_N(t) \in f(Q)$ for every t.

(H2) Regular lift. The resolved-rate solution $\dot{q} = J^{+}(q)\,\dot{\gamma}_N(t)$, started at any q_0 with $f(q_0) = \gamma_N(0)$, exists on $[0, 2\pi]$. It stays within joint limits, and $\sigma_{\min}(J(q(t))) \geq \sigma_0 > 0$ along it.

(H3) Bandwidth. When the loop is executed in time T, the controller tracks task-space references with content up to frequency f_c to within ε_C, and $N^*(\varepsilon_F) / T \leq f_c$.

Theorem 2. *Under (H1)–(H3) the pen traces γ to within ε. Kinematically the tracing is exact: $f(q(t)) = \gamma_N(t)$. The joint speeds are bounded by $\|\dot{q}\| \leq \|\dot{\gamma}_N\| / \sigma_0$, and the shortest admissible loop period is $T_{min} = N^*(\varepsilon_F) / f_c$.*

Proof. By (H2), J has full row rank along q(t), so $JJ^{+} = I$ and $\frac{d}{dt} f(q(t)) = J\dot{q} = \dot{\gamma}_N$. Since $f(q(0)) = \gamma_N(0)$, this gives $f(q(t)) = \gamma_N(t)$. The bound $\|J^{+}\| = 1/\sigma_{\min} \leq 1/\sigma_0$ gives the speed bound. Harmonic k of γ_N has frequency k/T, so (H3) puts every harmonic inside the tracked band and the executed curve is within ε_C of γ_N. By Theorem 1 that is within $\varepsilon_F + \varepsilon_C = \varepsilon$ of γ. ■

Caveat. (H3) is a requirement on task-space tracking, which is how operational-space controllers are specified. The joint trajectories q(t) are not band-limited even though γ_N is, because the kinematics are nonlinear. Version 1's sentence "and per the measurements, it does [suffice]" is withdrawn: the measurements below give $N^*(\varepsilon)$, which is one side of the inequality in (H3). The other side, f_c, has to be measured on the actuators.

1.5 A measurable dexterity index

The four-factor index from v1 is kept, with each factor redefined so that it is well defined and tied to Theorem 2. For a task curve Γ, tolerance ε and loop period T:

D1 Bandwidth sufficiency: $D1 = \min(1, f_c \cdot T / N^*(\varepsilon_F))$ — this is (H3) as a score.

D2 Manipulability along the path: $D2 = \min_t w(q(t)) / w_{ref}$, where $w = \sqrt{\det(JJ^{T})}$ (Yoshikawa) and w_{ref} is the maximum of w over the design's workspace (v1 left this reference undefined).

D3 Isotropy along the path: $D3 = \min_t \sigma_{\min}(J) / \sigma_{\max}(J)$; it is zero exactly when (H2) fails.

D4 Reach coverage: $D4 = \text{length}(\Gamma \cap W) / \text{length}(\Gamma)$. v1 used the volume of Γ, which is zero for any curve.

$$D = (D1 \cdot D2 \cdot D3 \cdot D4)^{\{1/4\}}$$

D = 0 whenever any factor is zero. In particular a hand that cannot reach its outline (D4 = 0) has no dexterity on this task however many joints it has — reach comes before dexterity, as v1 said.

Worked example (the bandwidth figure is assumed, not measured). With f_c = 5 Hz, tolerance ±1 mm and a 10 s loop, $D1 = \min(1, 50/61) = 0.82$. Stretching the loop to 12.2 s brings D1 to 1.

1.6 Self-inscription

The hand's outline depends on its pose and moves with the hand, so “drawing its own outline” has to be pinned down. Fix a reference pose q_ref (for instance, open palm flat on the canvas).

Def. ι(M) is the closed boundary curve of hand design M in pose q_ref: the outline, measured once and stored. τ(Γ) is the joint trajectory of Theorem 2 for curve Γ. The draw map D_M(Γ) is the curve that M actually traces when it runs τ(Γ).

Def. M is (s, ε)-self-inscribing if $\|D_M(s \cdot \iota(M)) - s \cdot \iota(M)\|_\infty \le \varepsilon$: the scaled outline is an ε-fixed point of M's own draw map. (v1's “τ ∘ ι = id” does not type-check, because τ returns joint trajectories and ι returns curves.)

Corollary 2. *If $s \cdot \iota(M)$ satisfies (H1)–(H3) for M at tolerance ε, then M is (s, ε)-self-inscribing. Since Fourier coefficients scale linearly with the curve, $N^*_{sΓ}(ε) = N^*_Γ(ε / s)$: drawing the hand at half size to ±1 mm costs the same harmonics as full size to ±2 mm (36, rather than 61).*

Proof. Theorem 2 applied to $Γ = s \cdot \iota(M)$; the scaling identity holds because $\hat{c}_k(sγ) = s\,\hat{c}_k(γ)$. ■

This is what the voice note was after: the program that draws the hand comes from the hand's own measured geometry, and the hand's mechanics carry that program out. The hand measures the outline once, in q_ref, and then draws it in a different pose, holding the pen. It does not trace its own moving boundary.

1.7 Conjecture 1 — from drawing to printing

Drawing a curve is not making a hand. Version 1's “manufacturing cascade” from a single master hand is therefore restated as an open conjecture. There is a concrete bridge, though. A layer-wise 3D print of a shell is a stack of closed contours, and each contour is a closed curve that Theorem 1 covers.

Conjecture 1. *If every layer contour of a hand's shell, at scale s, satisfies (H1)–(H3) for a print head carried by that hand, then the hand can print its own shell geometry at scale s to tolerance ε.*

Even if this is proved, it covers geometry only. Joints, actuators, sensing and electronics are outside it, so self-replication in the full sense remains open.

1.8 Measured results

Method. The outline is a five-finger adult hand silhouette built from a palm, a wrist and five rounded fingers, with every corner given a 3 mm fillet. Hand length (wrist crease to middle fingertip) is L = 189.4 mm, and the perimeter is 1056.3 mm. The curve is resampled at 8192 equal arc-length steps and its FFT coefficients are computed; the error reported is the sup of the Euclidean distance between the outline and the partial

sum S_N. The 3D curve is the same outline wrapped onto a cylinder of radius 90 mm across the palm, standing in for the transverse metacarpal arch; this is a modelling assumption. No smoothing is applied.



Figure 1.1 — Left: the hand outline and its Fourier reconstructions at N = 8, 26 and 61. Centre: sup error against N on log–log axes, with the O(N $^{-2}$) rate of Corollary 1 for reference. Right: the 3D curve on the palm arch.

N	2D error (mm)	3D error (mm)	3D error, % of L
1	63.675	64.419	34.01%
2	44.825	45.141	23.83%
4	43.781	43.764	23.11%
8	12.897	12.887	6.80%
12	5.729	5.999	3.17%
16	6.110	6.111	3.23%
26	3.125	3.075	1.62%
32	2.696	2.690	1.42%
48	1.437	1.449	0.77%
64	0.826	0.818	0.43%
100	0.264	0.264	0.14%
128	0.158	0.158	0.08%
200	0.063	0.063	0.03%
256	0.033	0.033	0.02%
400	0.010	0.011	0.01%

The error is not monotone in N (for example N = 12 gives a smaller error than N = 16), so N* is defined as the smallest N after which every larger N meets the tolerance. The minimum loop periods below use illustrative task-space bandwidths of 2, 5 and 10 Hz; these are assumptions to be replaced by measured actuator figures.

Tolerance	N* (2D = 3D)	Circles (2N)	T_min at 2 Hz	at 5 Hz	at 10 Hz
±2 mm (1.06% of L)	36	72	18.0 s	7.2 s	3.6 s
±1 mm (0.53% of L)	61	122	30.5 s	12.2 s	6.1 s
±0.5 mm (0.26% of L)	77	154	38.5 s	15.4 s	7.7 s
±0.25 mm (0.13% of L)	109	218	54.5 s	21.8 s	10.9 s

Comparison with v1. The v1 outline was not a hand: its own figures show a smooth loop with a small squiggle and no fingers. It was also passed through a periodic Gaussian smoothing (σ = 8 samples on a 512-point loop). That filter multiplies harmonic

16 by 0.29, harmonic 32 by 0.0072 and harmonic 48 by about 10^{-5}, which is why v1's error fell to zero by N = 48. Applied to the real hand, it shortens the fingers and fills the webs, moving the outline by up to 11.5 mm (Figure 1.2). On the real outline, the harmonics needed are 36 at ±2 mm (v1: 14) and 61 at ±1 mm (v1: 19) — about 2.6 to 3.2 times v1's figures.

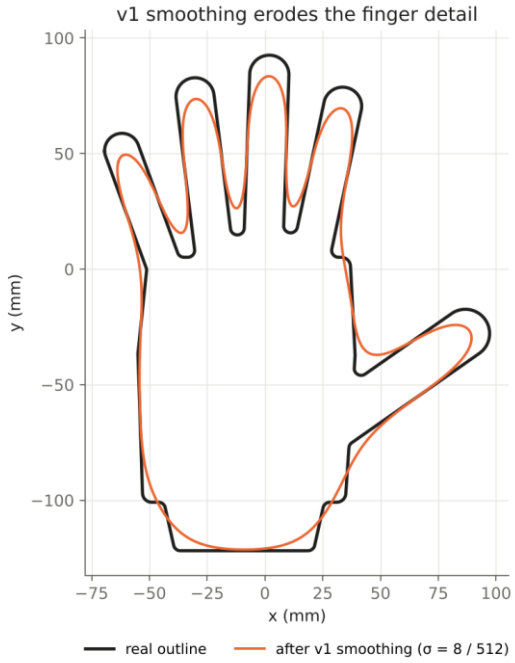


Figure 1.2 — The real outline (black) and the same outline after v1's smoothing (orange).

Companion file: epicycle_hand_v2.gif. It animates the chain of 122 counter-rotating circles (N = 61) drawing the hand to ±1 mm. Circles under 0.6 mm are not drawn.

1.9 Conclusion

The intuition in the voice note holds. Rotating circles, overlapping and stacked tip to tail, can draw any closed curve in R^3 (Theorem 1). In 3D each harmonic is a tilted ellipse made of two counter-rotating circles, with the chain held inside overlapping spheres (Lemma 1). A hand outline with rounded corners costs $O(N^{-2})$ error per harmonic budget, and the real outline needs 61 harmonics for ±1 mm.

The correction concerns what dexterity measures. A Fourier machine has a single degree of freedom, so joint count does not buy harmonics. A manipulator draws a given curve when three conditions hold: it can reach the curve, it stays away from singularities along it, and its task-space bandwidth covers $N^*(\varepsilon)/T$ (Theorem 2). Self-inscription is then a precise, testable property: the scaled outline is an ε-fixed point of the hand's own draw map (Corollary 2). Dexterity is still a bandwidth number, as v1 said, but it is a bandwidth in hertz measured against the harmonic content of the shape, not a count of joints.

Next steps: (1) choose a concrete kinematic model and check (H1)–(H2) numerically for $s \cdot \iota(M)$; (2) measure f_c on the chosen actuators to complete D1; (3) attempt Conjecture 1 on a single printed layer.

1.10 Changes from version 1

v1 statement	**Status**	**v2**
Ellipse semi-axes $(\|\alpha\| \pm \|\beta\|)/2$	Wrong (a unit circle gives 1 and 0)	Singular values σ_1, σ_2 of $[\alpha\ \beta]$ (Lemma 1)
Each ellipse lies on a sphere	Wrong unless it is a circle	Lies in the shell $\sigma_2 \le r \le \sigma_1$; overlapping bounding balls
Rate C/N^2 from $\|c_k\| = O(k^{-2})$	Inconsistent; that decay gives $O(1/N)$	$O(1/N)$ with corners, $O(N^{-2})$ with fillets; measured −1.06 and −2.35
27 DOF buys N = 8 (gimbal) or N = 26 (spin)	Withdrawn	A Fourier machine has 1 DOF; DOF gives reach and regularity, bandwidth gives harmonics
±2 mm met under the gimbal scheme	Contradicted v1's own table (3.87 > 2)	Superseded by the N* table
Thumb IP 2 DOF; palm arch; forearm lock-out	Non-standard	16 + 5 + 6 = 27 (§1.3)
Measured on a smoothed outline	Not a hand; the smoothing set the result	Five-finger outline, no smoothing (§1.8)
D4 = Vol(W ∩ Γ) / Vol(Γ)	Zero for any curve	Arc-length ratio
$\tau \circ \iota = \mathrm{id}$	Does not type-check	ε-fixed point of the draw map (§1.6)
Master hand prints copies	Unproven	Conjecture 1, geometry only